



Quantum-based predictive fog scheduler for IoT applications

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ABSTRACT

Load scheduling across distributed fog computing nodes has been a major challenge to meet the increased demand of real-time data analysis, and time-sensitive decision-making. This study presents a Quantum Computing-inspired (QCI) optimized load scheduling technique in fog computing environments for real-time IoT applications. In addition to this, QCI-Neural Network Model is used as a predictive model to determine the optimal computational node for enabling real-time service delivery. For validation, simulations were performed using 3, 5, 7, and 10 fog nodes at different instances to schedule nearly 600 heterogeneous tasks to obtain the respective results. Results were compared with other state-of-the-art scheduling models like Min-Max, Minimum Completion time, and Round Robin. Better results were registered for the proposed technique in terms of minimum average task completion time (28.15 s for 10 nodes) and average energy consumption (3.48 J for 10 nodes). Moreover, higher rates of statistical parameters like sensitivity (88.66%), specificity (91.28%), precision (92.25%), and coverage (95.66%) were acquired depicting enhanced overall performance.

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1. Introduction

Innovations in Internet of Things (IoT) technology have revolutionized industrial sectors significantly leading to the notion of Industry 4.0 [1,2]. Smart Health, Smart Education, and Smart Transportation are few examples where IoT has been significantly productive [3]. In fact, according to a report, global industrial IoT is expected to reach 91.40 billion dollars by the year 2023.¹ Consequently, real-time service provisioning, and automation of decision-making have become important areas of research in modern era [4]. Increased utilization of IoT devices have raised the demand of computational power for effective data analysis with minimal delay [5]. However, due to limitations of energy-computing-bandwidth, till date, it becomes difficult to meet these challenges with conventional techniques. The forthcoming internet revolution has introduced a new notion of fog computing. Fog computing is a novel computing paradigm that aims at extending the cloud computing services to the access network, in order to minimize delays induced by service deployments [6]. Fog-based resource scheduling comprises of distributing interconnected

computational nodes to provide real-time services in an efficient manner [7]. Even though many definitions of fog computing have been put forth by researchers around the world, this paper provides a comprehensive definition as discussed ahead.

Definition 1. Fog computing is defined as a virtualized platform for provisioning real-time computation, data storage, and services at the network between the users and the cloud data centers, which are not located at the network edge.

Definition 1 provides a comprehensive overview of the innovative fog computing paradigm, capable of provisioning real-time services to the underneath IoT technology. A generic fog computing architecture has been shown in Fig. 1 depicting the utilization of fog layer for industrial IoT services.

1.1. Research motivation: applicability perspective

Fog computing acts as an important player for allocating heterogeneous resources dynamically in almost all industrial sectors. As categorized by Aazam et al. [8], modern smart industry is classified into three sectors, namely Primary (extraction related like mining), Secondary (manufacturing related like automobiles), and Tertiary (service related like transportation). With potential to become a part of Industrial IoT vision, fog computing has uncovered new directions to achieve goals of futuristic smart industry. Some of the important industrial areas where fog-based computation can enhance services include

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¹ Source: <https://www.marketsandmarkets.com/Market-Reports/industrial-internet-of-things-market-129733727.html>.